NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI - 620015

Instrumentation and Control Engineering

CEIR11 – Basics of Civil Engineering

Assignment 1

Semester: I **Session**: July 2026

Branch: B. Tech. (ICE) **Deadline**: 11/11/2026

1. Explain the role of civil engineering in the planning and construction of industrial buildings, instrumentation laboratories, and control rooms. Discuss the selection of suitable building materials and foundations with relevant examples.
2. Prepare a comparative chart showing the properties, uses, and environmental impacts of OPC, PPC, and PSC cements. Suggest the most suitable cement type for residential and marine construction, with reasons.
3. Explain the various shapes and sections of steel used in building works, and describe their engineering applications.
4. Explain in detail the various types of bonds in brick masonry with neat sketches.
5. Select a suitable site within your locality for constructing a residential building. Evaluate its suitability based on topography, soil condition, drainage, accessibility, and availability of utilities. Support your answer with a sketch map and short justification.
6. Draw a labelled sketch showing the main components of a building (foundation to roof). Explain the function of each component and suggest suitable materials for construction.
7. Prepare a brief case study on a building or structure that experienced foundation settlement or failure. Identify the possible causes, consequences, and suggest remedial or preventive measures to avoid such failures in future projects.
8. Develop a conceptual plan for a small single-storey residential building (2BHK or the equivalent of your home, preferably). Choose suitable materials for each component (foundation, wall, roof, etc.), specify the foundation type based on assumed soil condition, and justify your choices through engineering reasoning.